
```

% EEL3135 Lab 3
% Name: Evan Baesler
% UFID: 31151619

% Part 1 -
% Calculate the phasors using Matlab in polar form for each set of Fourier
coefficients.
%The code should use a for-loop where the amplitude and phase are held in
the vectors, Amp and Phi (provide
%the code and solution).
%Amp = [A-9, A-8, A-7, A-6, A-5, A-4, A-3, A-2, A-1, A0, A1, A2, A3, A4, A5,
A6, A7, A8, A9]
%Phi = [ $\phi$ -9,  $\phi$ -8,  $\phi$ -7,  $\phi$ -6,  $\phi$ -5,  $\phi$ -4,  $\phi$ -3,  $\phi$ -2,  $\phi$ -1,  $\phi$ 0,  $\phi$ 1,  $\phi$ 2,  $\phi$ 3,  $\phi$ 4,  $\phi$ 5,
 $\phi$ 6,  $\phi$ 7,  $\phi$ 8,  $\phi$ 9]

data = [3-2.5*i, 0+0*i, 2-2.5*i, 0+0*i, 0+0*i, 3-3*i, 0+0*i, 8-5*i, 0+0*i,
5, 0+0*i, 8+5*i, 0+0*i, 3+3*i, 2+2.5*i, 0+0*i, 2+2.5*i, 0+0*i, 3+2.5*i];

% Taking length of data, and setting up 1x19 arrays to store each waveforms
% amplitude and phase, along with constructing the equivalent cosine wave.
N = length(data);
Phasor = zeros(1, N);
Amp = zeros(1,N);
Phi = zeros(1,N);

% Uses data to find amplitude and phase, storing it and constructing our
% waveforms.
for count = 1:N
    Amp(count) = abs(data(count));
    Phi(count) = angle(data(count));
    Phasor(count) = Amp(count) * exp(1j * Phi(count));
end

% Part 2 -
% Provide the MatLab code for the time vector, "t" that will be used in
% the Fourier synthesis equation. This will be held as a vector and used
% to build your sinusoid components.

fs = 80; % Given by prompt.
dt = 1/fs; % Sampling rate is 1/fs
t = 0:dt:1-dt; % Time is set to give 80 samples within a 1 second period.

% Part 3 -
% Create another for loop that will loop through the ak fourier
% coefficients which will create each sinusoidal component Xk(t).
% These sinusoidal components will be held in a Matrix, M(K, N),
% where k is the number of fourier coefficients and N is the length of
% the signal.

N = length(t); % Makes N a 1x80 array
K = 19; % We have 19 data points. (-9 to 9)
M = zeros(K,N); % Makes an array of 19x80 (19 waves, 80 data points)

```

```

f0 = 4; % Fundamental frequency for 4 Hz steps

for I = 1:19
    % Finds ak by finding equivalent phasor stored
    ak = Phasor(I);
    % Finds real values of each point in our wave
    M(I, :) = real(ak * exp(1j * 2 * pi * I * f0 * t));
end

% Part 4 -
% Provide the code for how the Fourier synthesis will be implemented.

% Sums all waves together
netWave = sum(M,1);

figure;
plot(t, netWave)

% Part 5 -
% Plot 3 rows of the Matrix that are tied to an ak. Describe the signals
% for each plot and its relationship to the components of a sinusoidal wave.

% Plots wave a_-9
figure;
plot(t, M(1,:));
title('a_-9 component')
xlabel('Time (s)')
ylabel('Amplitude')

% Plots wave a_0
figure;
plot(t, M(10,:));
title('a_0 component')
xlabel('Time (s)')
ylabel('Amplitude')

% Plots wave a_9
figure;
plot(t, M(19,:));
title('a_9 component')
xlabel('Time (s)')
ylabel('Amplitude')

% Part 6 -
% Add the Matrix rows to then construct the X(t)Sum signal, using the
% matlab function "sum". X_t_sum

X_t_sum = sum(M, 1);

% Part 7 -
% What is the DC offset of the synthesized signal and how does this relate
% to the fourier coefficients?

% a_0 is our phasor at 0 Hz which is our DC offset (real component):

```

```
DC_offset = real(Phasor(10))

% DC_offset = +5V

% Part 8 -
% Provide a plot of the synthesized signal. Use the grid on feature on
% the plot and describe what the graph represents.

figure;
plot(t, X_t_sum);
title('Combined Fourier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

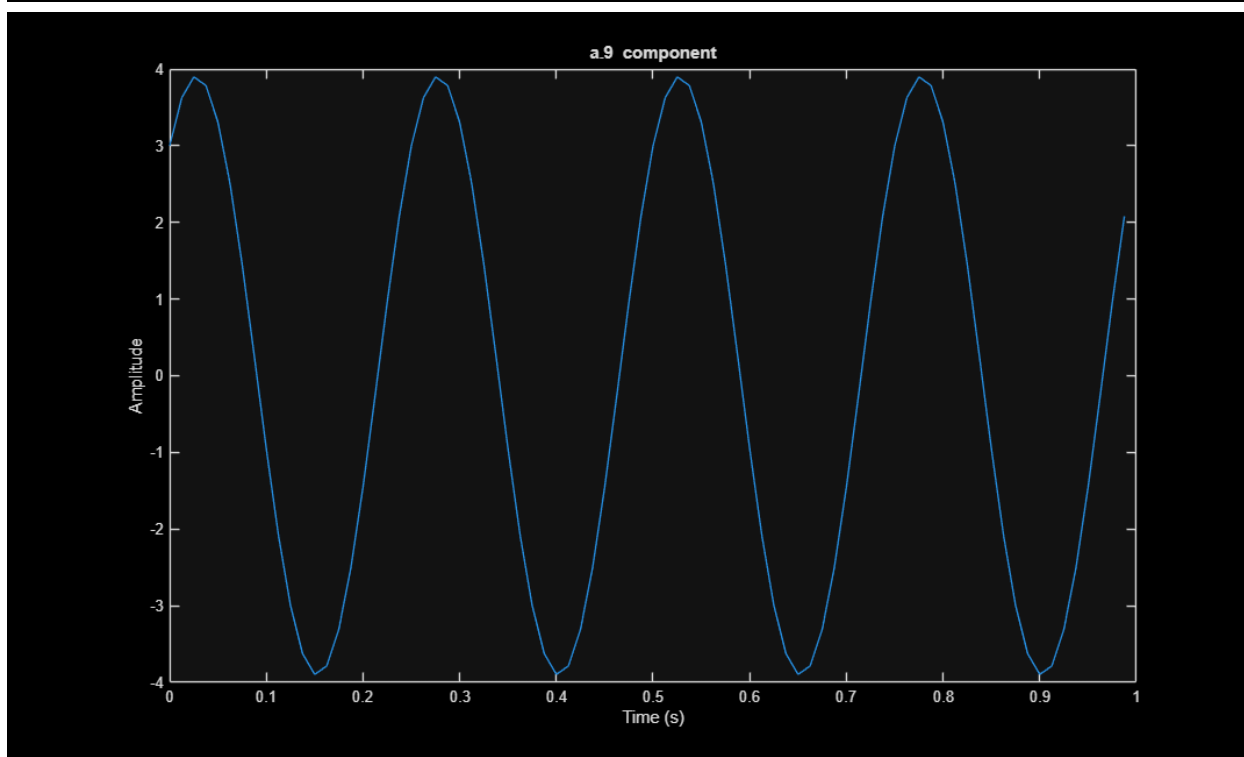
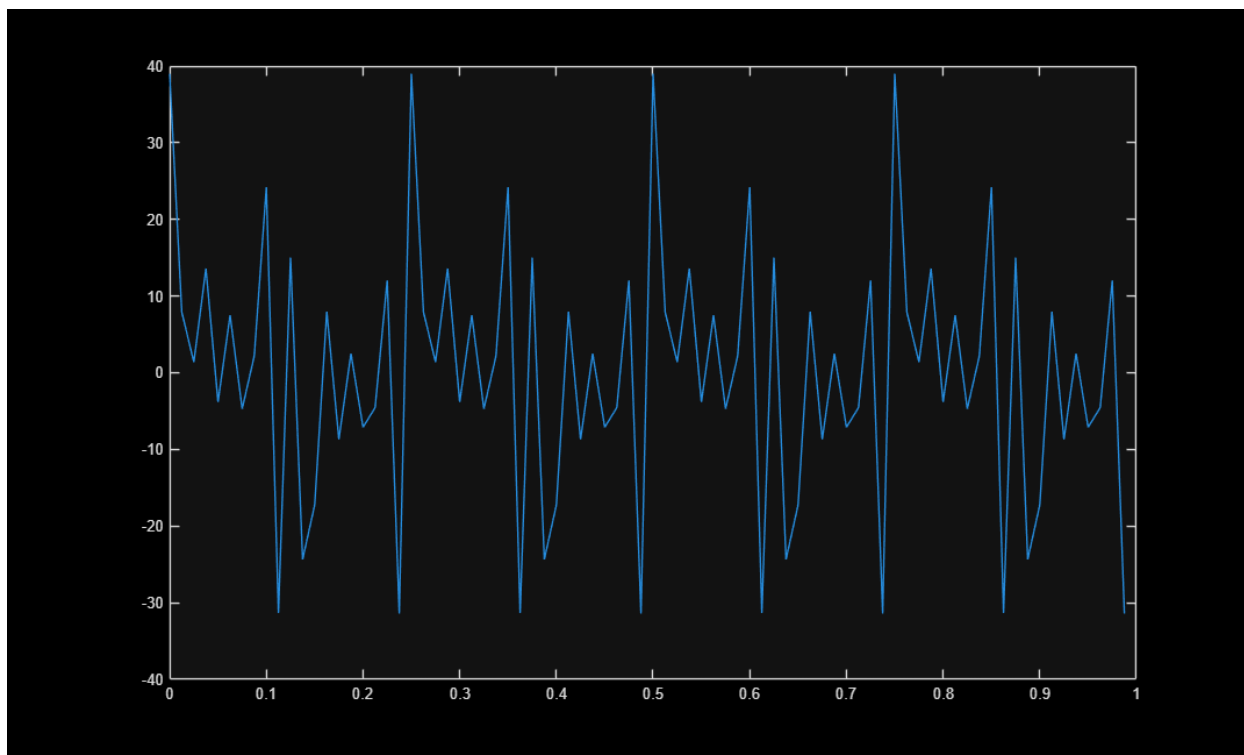
% The signal represents the secret message sent by the mainland formed by
% their compressed sequence sent by signals that step by 4 Hz values,
% going as low as -36 Hz to +36 Hz, from a_-9 to a_9.

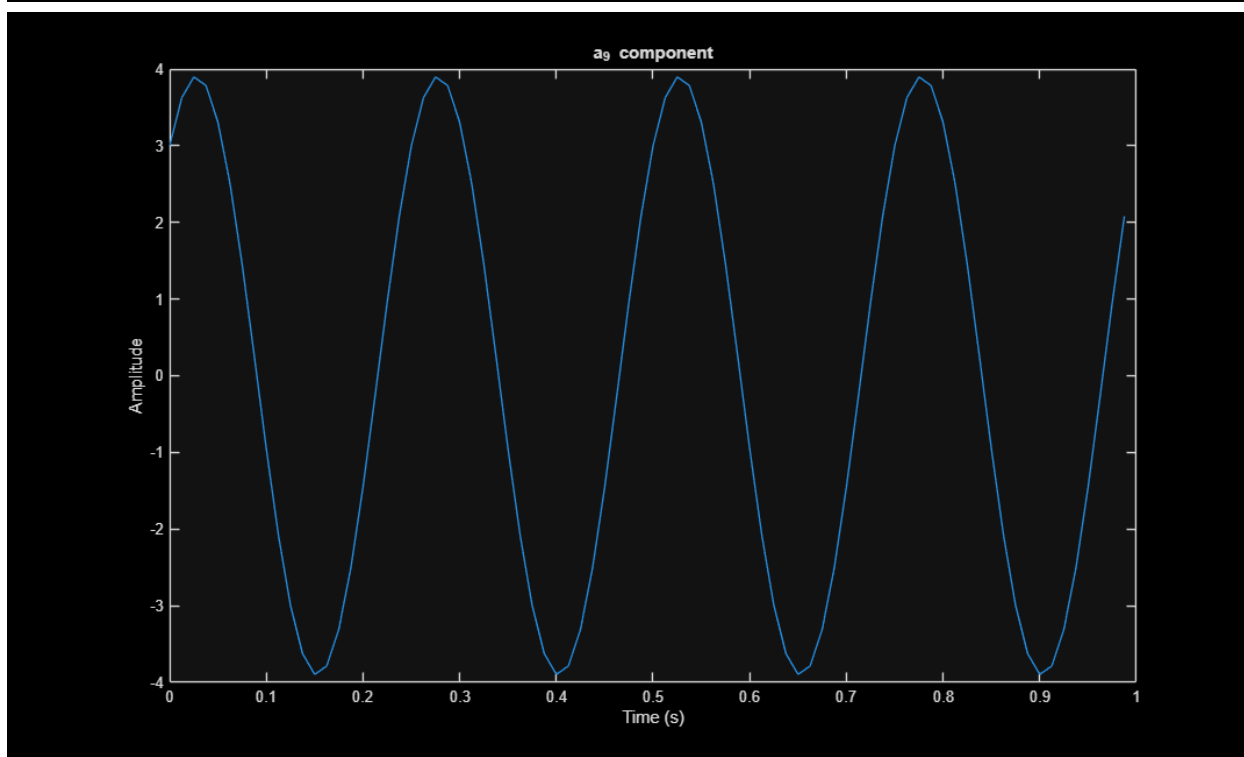
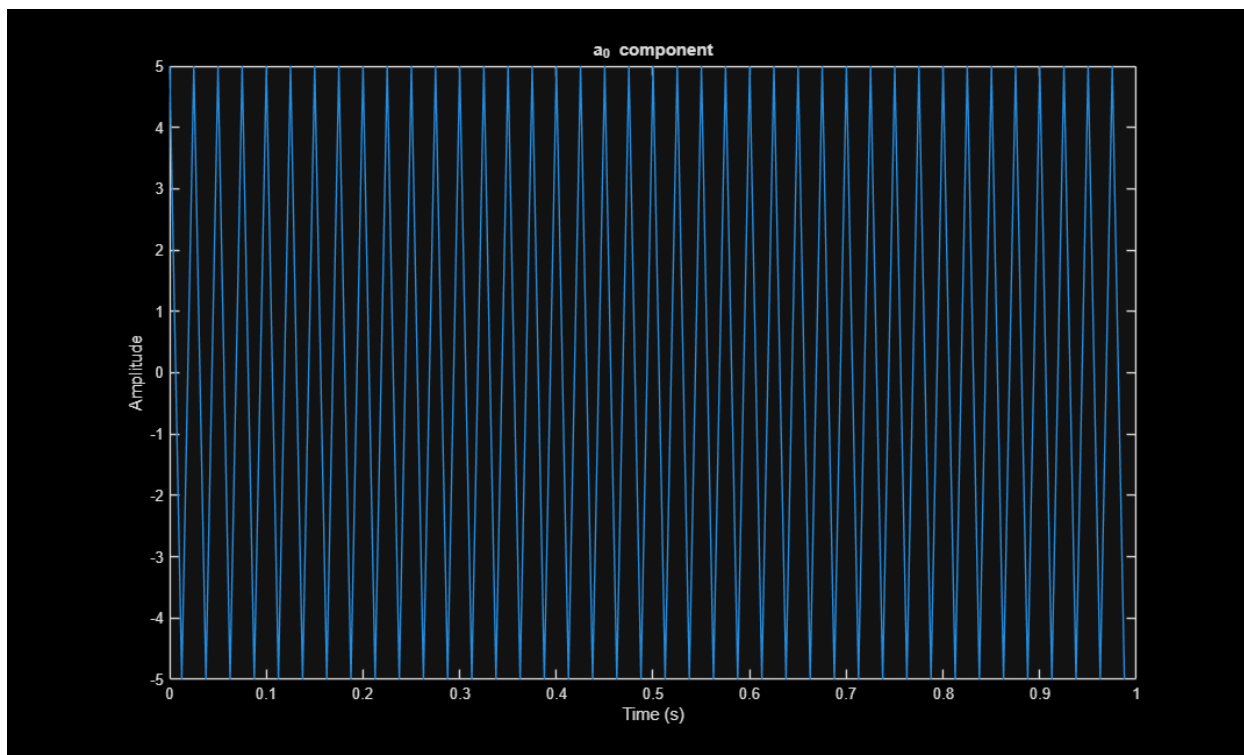
% The problem gave us 19 possible harmonics, and 9 phasors in rectangular
% form, which we used to find amplitude and phase, then we formed the
% equivalent waveforms, later combining them into one singular waveform.

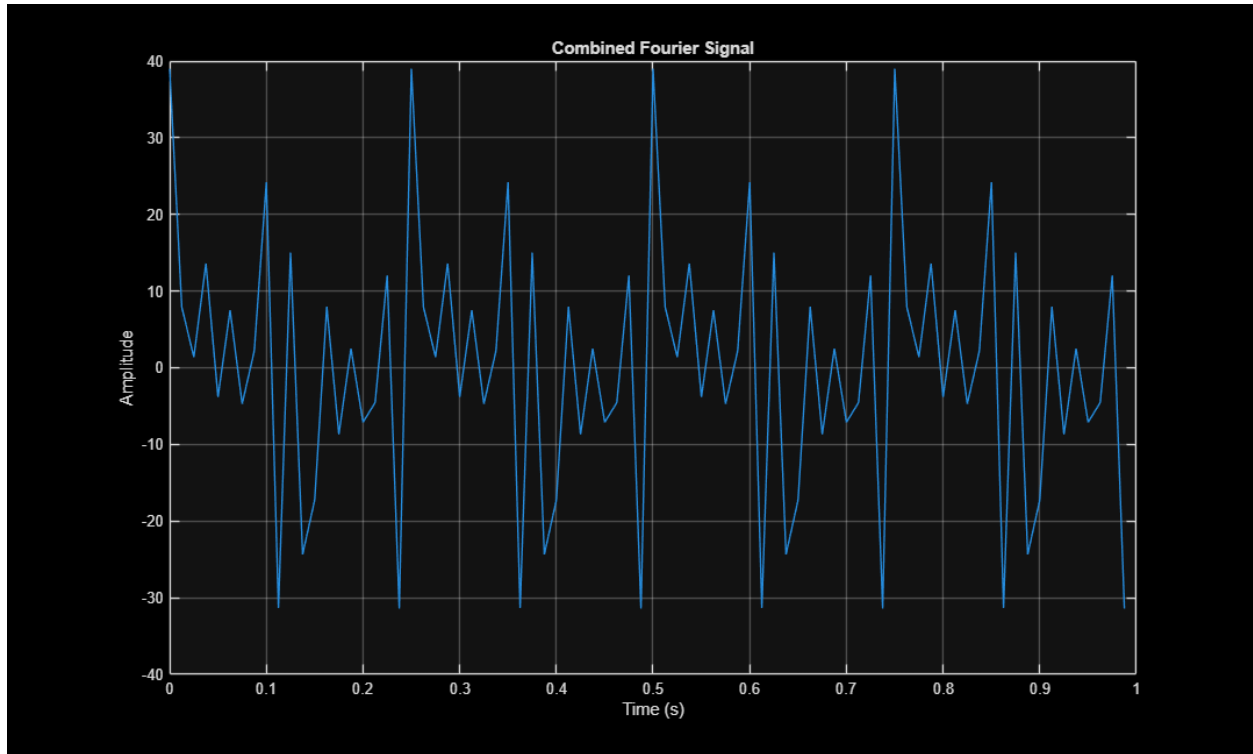
% This is a fourier synthesis, where we were given simple cosine waves and
% combined them together to form an advanced signal.

DC_offset =

5
```







Published with MATLAB® R2025a